Teaching demonstration

Understanding Inflation

Why Prices Rise and How Central Banks Respond

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Three questions for today

1. What is inflation and how is it measured?
2. Is inflation good or bad?
3. What do central banks do, and at what cost?

A receipt from the same store, 24 years apart



Item	2002	2026	Change
Rice, 5 kg	¥980	¥4,200	+329%
Milk, 1 L	¥98	¥260	+165%
Eggs, 10	¥78	¥310	+297%
Bread, 1 loaf	¥68	¥210	+209%
Cabbage	¥58	¥230	+297%
Total	**¥1,282**	**¥5,210**	**+306%**

Store: JUSCO Hadano, 12 May 2002 → AEON Hadano, 12 Oct 2026

(same supermarket; JUSCO was later renamed AEON)

Average of the five product increases: $(329 + 165 + 297 + 209 + 297) / 5 = \mathbf{259\%}$

Average inflation per year (compound): $\left(\frac{\text{final}}{\text{initial}}\right)^{1/\text{years}} - 1 = \left(\frac{5210}{1282}\right)^{1/24} - 1 \approx \mathbf{6\%}$

From the receipt to the inflation rate

Do you think 6% is the inflation rate?

Not everyone buys the same things.
So how should the items be weighted?

The Consumer Price Index

1. Definition

The cost of a fixed basket of goods and services, relative to a base period.

2. Weights

Weight = category spending ÷ total spending.

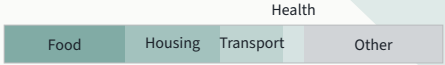
Larger share of spending ⇒ larger weight.

3. Formula

$$\mathrm{CPI}_t = 100 \times \frac{\sum p_t q_0}{\sum p_0 q_0}$$

p = price · q = quantity · 0 = base

What's in the basket

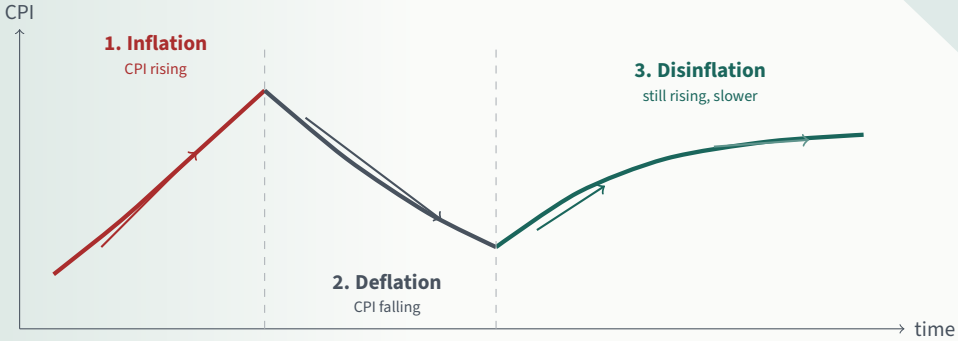


Food	27.5%
Housing	21.8%
Transport	14.4%
Health	4.7%
Other	31.5%

Statistics Bureau of Japan, 2025 base

Inflation, Deflation, and Disinflation

Vertical axis = CPI (price level). Slope of the curve = inflation.



Disinflation is last: the level is still up, but the rise is slowing.